2022 7th International Conference on Intelligent Computing and Signal Processing (ICSP) | 978-1-6654-7857-1/22/$31.00 ©2022 IEEE | DOI: 10.1109/ICSP54964.2022.9778786

Motor Imagery Tasks EEG Signals Classification Using ResNet with Multi-Time-Frequency Representation

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***Abstract*—This paper presents a ResNet-based multi-feature fusion method for motor imagery task classification in brain-computer interfaces. To address the drawbacks of insufficient information and low signal-to-noise ratio extracted from EEG signal features by a single time-frequency transform. Three time-frequency transforms including morlet wavelet transform, multi-window time-frequency transform and stockwell transform are used to extract features from EEG signals. Those features are then input to three ResNets for further feature extraction, and finally the high-level feature outputs by the three networks are fused before the fully connected layer of the original ResNet, and then fed into a fully connected layer for classification. The performance of the proposed method is evaluated using the accuracy and kappa value on the IVa dataset of BCI Competition III, and the results show that the ResNet model using feature fusion produces better results with a classification accuracy of 97.86%, which is 39.65% higher than the model using only one time-frequency transform as input.**

Keywords-MI; EEG; ResNet; BCI

I. INTRODUCTION

Brain-computer interface (BCI) plays a crucial role as an information pathway between the human brain and the outside world, especially in the newly proposed metaverse concept where immersive interactive experience is of paramount importance and where the BCI is the perfect solution for noninvasive connectivity. Motor imagery refers to EEG signals that imagine body movement without actual movement. When subjects imagine body parts moving, the energy of motor imagery signals will decrease or increase in the frequency range of mu waves (8-12Hz) and beta waves (18-26Hz). EEG signals with such characteristics are called events Correlation Desynchronization (ERD) or Event Related Synchronization (ERS). Therefore, motor imagery signals have significant time-frequency characteristics. In addition, there are differences in the brain regions activated by different motor imagery classification tasks, that is, motor imagery signals also have spatial distribution characteristics. The key to extracting and identifying the EEG signal of this feature is to effectively utilize the time, frequency and space information of the motor imagery EEG signal. In recent years, with the development of deep learning technology, many researchers have applied it to EEG signal classification. In the field of computer vision, convolutional neural network is an important representative of deep learning. With its end-to-end structure, In the process of extracting EEG signal features using CNN, the interference of EEG noise information can be avoided, thereby increasing the accuracy of classification tasks. However, due to the complex components of EEG signal components, the accuracy of the current direct feature extraction method using CNN is relatively low. As images are translation invariance, its spatial properties fully satisfy the requirements of CNN convolution operations, while EEG signals are nonstationary time-series signals that cannot be directly used in CNN, and usually use the morlet wavelet transform to convert the time-series to the time-frequency domain before convolution. Tabar et al. used CNNs to extract time-frequency features that have undergone a short-time fourier transform to decode EEG signals [1]. Huo et al. obtained two-dimensional features with high time-frequency resolution by integrating the EEG source imaging method with the continuous wavelet transform method, and then used CNN to decode the EEG signal [2]. Dai et al. proposed the use of a mixed convolutional scale CNN model for decoding motor imagery EEG signals, mainly using convolution kernels of different sizes to extract the temporal and spatial features of EEG signals. [3]. The deeper the level of the neural network, the lower the training accuracy and the phenomenon of network degradation occurs [4]. Residual network proposes a residual block to reduce degradation phenomenon. Based on this, this paper proposes a ResNet motion imagery classification model that receives multiple time-frequency domain information inputs. Experiments demonstrate that multiple time-frequency domain features and the residual network effectively improve the classification accuracy of motion imagery.

II. METHOD

Fig.1 illustrates a motion imagery classification model based on ResNet with multiple time-frequency domain feature inputs. The original EEG signal is transformed into a 2D image by three time-frequency transforms (morlet wavelet transform, multi-window transform and stockwell-transform), then the images extracted by each of the three methods are input to three residual networks, and finally the advanced features extracted by convolution are combined and input to the fully connected layer for classification.

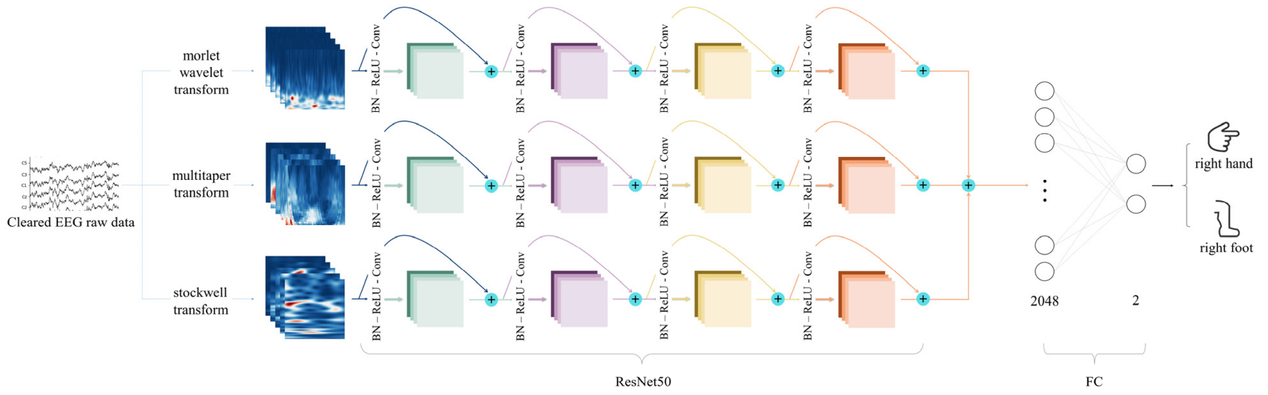
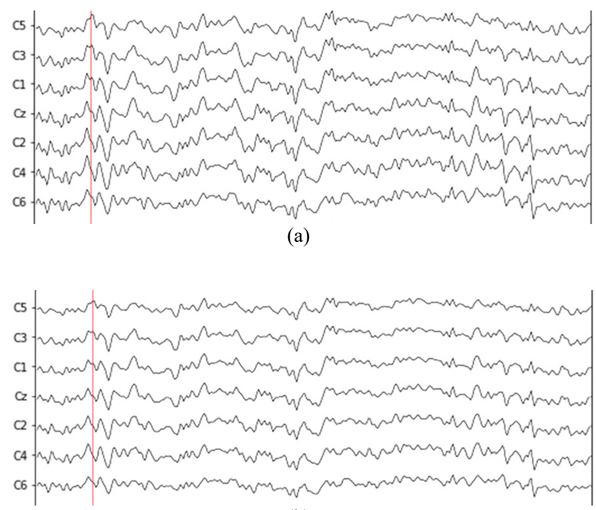


Figure 1. Structure diagram of ResNet-based motor imagery classification network

A. DateSet

This paper is based on dataset IVa of the BCI Competition-III [5]. Five healthy subjects were recorded performing right hand imagery and right foot imagery tasks. Using 118 electrodes from the international 10/20 system, motor imagery-related EEG information was acquired. A 3.5-second visual cue was presented to the subjects to activate their motor imagery during the experiment. A total of 280 trials were recorded for each subject, 140 of which were for each type of task. The data from individual subjects were divided according to a ratio of 80% training set and 20% test set.



(a)

(b)

Figure 2. EEG signals before (a) and after (b) preprocessing

B. Preprocessing

The raw EEG signal was sampled at 1000Hz, and the EEG waveform recorded in one motor imagery experiment is shown in Fig.2 (a). Then a band-pass filtering of 0.5-50Hz was performed and down-sampled to 100Hz, and then FastICA was used to remove artifacts and eye movements from the EEG signal. The EEG waveform after cleaning up the noise is shown in Fig.2 (b). Upon observation, it can be seen that the pre-processed waveform is much smoother. For example, a peak at 0.35s is reduced after ICA to remove blink artefacts. Details of the dataset used can be found in [5].

C. Time-Frequency Representations

In the field of mathematics, the morlet wavelet [6] is wavelets consisting of complex indices (carriers) multiplied by Gaussian windows (envelopes), which are intimately associated with human auditory and visual perception.

The morlet wavelet transform is suitable for analyzing signals that are not stationary and has been used in medicine to distinguish abnormal heartbeat behavior. The EEG signal is also a nonstationary signal, so the morlet wavelet transform was chosen as the method for extracting features from the EEG signal.

The multi-window time-frequency transform [7] is a technique developed by David J. Thomson that overcomes some limitations of the nonparametric fourier transform. When the number of samples is small but mixed with a large amount of noise, the noise can be reduced by averaging. However, it also weakens the signal components that vary from experiment to experiment, and using this method is not desirable. Instead, the multi-window time-frequency transform avoids these problems by sampling the same sample multiple times with different window sizes to obtain multiple independent estimates.

The stockwell transform [8], like other time-frequency analysis tools, allows us to look at the energy distribution of a signal in both the time and frequency domains. stockwell transform is unique in that it maintains a direct relationship with the fourier transform, while allowing different resolutions at different frequencies. At low frequencies the signal changes slowly and has a low time domain resolution and a high frequency domain resolution, while at high frequencies the signal changes rapidly and has a high time domain resolution and a low frequency domain resolution. The stockwell transform is also a good time-frequency transformation (TFR) method for EEG signals because the EEG signal varies between calm and stimulated states.

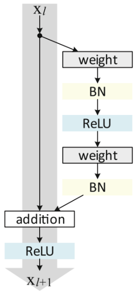


Figure 3. The structure of the residual block

D. Residual Block

ResNet consists of a series of residual blocks with different convolution parameters. Direct mapping and residual are the two sections of the residual block. The following is a description of a residual block:

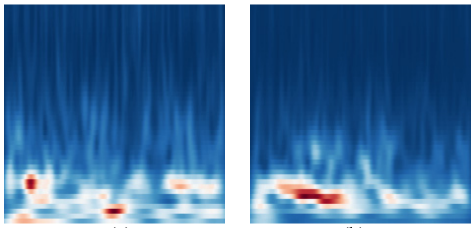
$$x_{l+1} = x_l + \mathcal{F}(x_l + W_l) \tag{1}$$

Among them, x_l is the direct mapping, i.e. the path on the left in Fig.3; $\mathcal{F}(x_l + W_l)$ is the residual portion, which usually comprises of 2 or 3 convolution processes, i.e. the path on the right in Fig.3; and W_l is the parameter of the convolution kernel. When the residual is zero, the residual block becomes a simple feature mapping, ensuring that the performance of the network does not degrade. However, the residual do not become completely zero, allowing the residual block to learn new properties based on input, hence strengthening the network's performance [9].

III. Results and Discussion

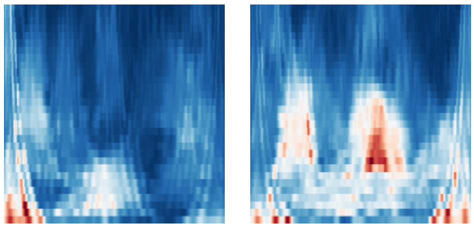
In this paper, three TFR methods, namely the morlet wavelet transform, the multi-window transform and the stockwell-transform, are utilized to extract features from the raw EEG signal in order to get more information for the motor imagery identification task. In this paper, the effectiveness of using three TFR methods and only one TFR method on a motor imagery classification task was investigated.

Firstly, the original EEG signal was transformed by three methods: morlet wavelet transform, multi-window transform and stockwell-transform. The frequency range is 5Hz to 50 Hz, where the window size of the morlet wavelet transform and the multi-window transform varied with frequency as 7/freq. The obtained spectrograms are shown in Fig.4 and Fig.5 respectively. The stockwell transform, on the other hand, utilizes a parameter to control the size of the Gaussian window, increasing the temporal resolution if the parameter is less than 1, and increasing the frequency resolution if the parameter is greater than 1. Since the ERD and ERS stimulated by the motion imagery are transient, and its frequency bandwidth is at least 4Hz, so the window parameter in this paper is taken as 0.65 to increase the temporal resolution of the spectrogram. stockwell-transform obtained the spectrograms as shown in Fig.6. Observing the results of the three time-frequency transforms, it can be concluded that the features extracted by the three time-frequency transforms are different, but the peaks can be found around the stimulation time points.



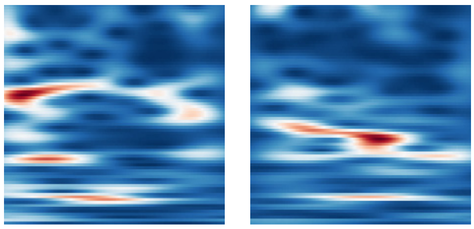
(a) (b)

Figure 4. Spectrograms of (a) right hand and (b) right foot obtained using morlet wavelet transform



(a) (b)

Figure 5. Spectrograms of (a) right hand and (b) right foot obtained using multi-window time-frequency transform.



(a) (b)

Figure 6. Spectrograms of (a) right hand and (b) right foot obtained using stockwell transform

As mentioned above, all three TFR images are resized to 224x224x3 and are used as input to the ResNet. Since three ResNet are used at the same time and only one output is required for classification, this paper considers merging the features extracted from the three networks before the fully connected layer of ResNet and feeding them into the fully connected layer for classification. As shown in Fig.7, after 400 epochs of training, the loss function converged with an accuracy of 99.64%. batch_size is set to 32, 9 iterations are performed per epoch, and evaluation is performed at each epoch. A stochastic gradient descent optimizer is used in the training with an initial learning rate of 0.001. Experiments are visualized using visualdl, which is used to show the loss values and accuracy rates during training.

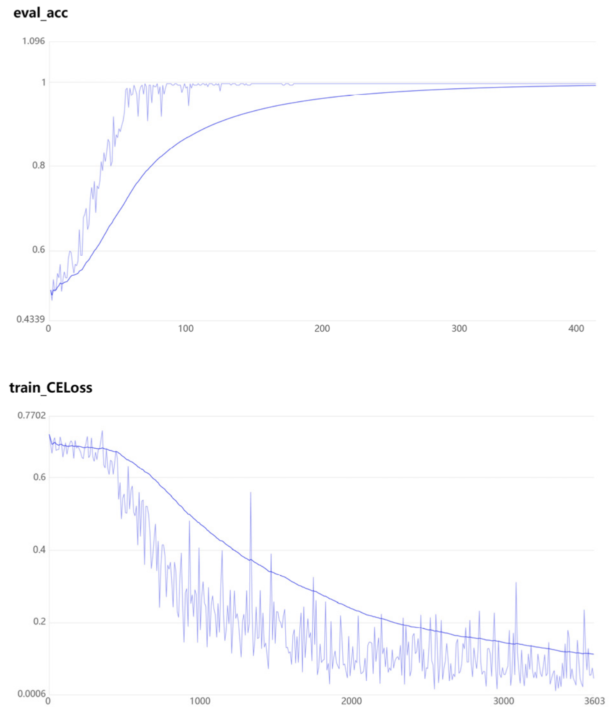


Figure 7. The training process of training the ResNet model using three time-frequency transformed result images

In this experiment, the accuracy and Kappa value were used as the indicators to evaluate the model. Its calculation formula is as follows:

$$Accuracy(\%) = \frac{TP + TN}{TP + TN + FP + FN} \times 100 \tag{2}$$

$$Kappa(\%) = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \times 100 \tag{3}$$

The confusion matrices generated using the three TFR approaches and utilizing only one TFR are shown in Table I. Table II shows the classification performance of the two approaches for these confusion matrices.

TABLE I. CONFUSION MATRIX

MI-Task	Only One TFR		Three TFRs	
	RH	RF	RH	RF
RH	TP=40	FP=100	TP=140	FP=0
RF	FN=17	TN=123	FN=6	TN=134

The total number of correctly detected positive and negative events is represented by TP and TN. In this paper, TP indicates that the actual classification labels are all right-handed, FP indicates that the actual label is right-handed and the model predicts that the label is right-foot, FN indicates that the actual label is right-foot and the predicted label is right-hand, and TN indicates that the actual classification labels and predicted labels are both right-hand and right-foot. It can be seen that on the current dataset, the model proposed in this paper classifies all the types labelled as right hand correctly, while for the right-foot type there are six classification errors. The model can be made to increase its focus on the difficult samples by data augmentation, adjusting the loss function and the regularization, which makes the difficult samples contribute more to the loss function and the model performs better. Table II shows that employing three TFR methods yielded better outcomes than using only one TFR approach. As a result, for the motion imagery task of discriminating between the right-hand and right-foot, the model using three TFR approaches is more accurate and effective.

TABLE II. CLASSIFICATION PERFORMANCE OF BOTH TFR METHODS

Parameters	Only One TFR	Three TFRs
Accuracy (%)	58.21	97.86
Kappa (%)	20.40	95.80

IV. CONCLUSION

This paper is based on ResNet and a multi-TFR approach for a common two-category motion imagery task in brain-computer interfaces. Inputting spectrograms that have undergone three separate TFR transformations into the three ResNets significantly improves the accuracy of motor imagery classification compared to a model using only one single TFR spectrogram input. The ResNet part in this study can also be optimized, and subsequent studies can combine the spectrograms obtained from the three TFRs into a single input, reducing the time and space required for training.

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